

COGNAC-CHATEAUBERNARD AB, France

WMO: 074120

Lat: 45.6650N

Lon: 0.3158W

Elev: 31

StdP: 100.95

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.8	(c) -2.0	(d) -7.4	(e) 2.0	(f) -1.3	(g) -5.6	(h) 2.4	(i) 0.1	(j) 12.5	(k) 11.7	(l) 11.2	(m) 11.6	(n) 2.4	(o) 40	(p) 0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.0	(c) 32.8	(d) 21.1	(e) 30.8	(f) 20.2	(g) 28.9	(h) 19.5	(i) 22.4	(j) 30.2	(k) 21.4	(l) 28.5	(m) 20.5	(n) 26.9	(o) 3.5	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 20.0	(b) 14.7	(c) 25.2	(d) 19.1	(e) 13.9	(f) 24.1	(g) 18.2	(h) 13.2	(i) 23.2	(j) 65.9	(k) 30.5	(l) 62.3	(m) 28.7	(n) 58.9	(o) 27.0	(p) 26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.2	8.9	7.9	-6.4	35.9	2.1	1.5	-7.9	37.0	-9.1	37.9	-10.3	38.8	-11.8	39.8
			-6.8	24.1	2.1	0.9	-8.4	24.7	-9.6	25.2	-10.8	25.7	-12.3	26.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.5	6.3	7.0	9.9	12.2	15.8	19.4	21.1	21.1	17.9	14.7	9.7	6.8
	DBStd	6.35	3.82	3.80	3.27	3.22	3.33	3.42	3.11	3.03	3.02	3.53	3.56	3.97
	HDD10.0	433	121	93	42	13	1	0	0	0	0	6	47	110
	HDD18.3	2049	372	316	262	184	94	27	9	7	43	121	259	356
	CDD10.0	1724	7	10	39	80	181	281	344	343	238	151	38	12
	CDD18.3	298	0	0	0	2	15	58	95	91	31	7	0	0
	CDH23.3	2848	0	0	2	33	168	580	889	855	285	36	0	0
	CDH26.7	987	0	0	0	3	37	202	331	336	75	3	0	0
Wind	WSAvg	3.5	3.9	4.0	4.1	3.8	3.5	3.3	3.3	3.0	3.0	3.3	3.6	3.7
Precipitation	PrecAvg	838	84	63	54	70	64	59	49	57	66	86	100	87
	PrecMax	1121	201	124	165	194	123	113	93	165	170	171	283	193
	PrecMin	631	15	2	4	11	7	3	14	11	6	26	22	15
	PrecStd	109	48	36	39	44	28	27	20	39	44	41	57	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.0	18.5	23.0	26.8	30.2	34.4	35.1	35.7	31.6	26.7	20.0	15.7
		MCWB	12.1	11.9	13.4	16.3	18.8	22.0	21.3	21.9	19.7	18.4	15.3	12.8
	2%	DB	13.7	15.7	20.2	23.7	27.4	31.1	32.3	33.0	28.9	24.0	17.7	14.3
		MCWB	11.8	10.8	12.7	14.8	17.8	20.4	20.8	21.3	18.8	17.5	14.3	12.4
	5%	DB	12.7	13.7	17.8	21.2	25.1	28.9	30.4	30.3	26.6	21.9	16.1	13.4
		MCWB	11.1	10.2	11.8	13.7	16.9	19.6	20.2	20.3	18.1	16.9	13.7	11.8
	10%	DB	11.6	12.2	15.7	18.8	22.8	26.7	28.2	28.0	24.5	20.1	14.9	12.4
		MCWB	10.1	10.1	10.8	12.5	16.0	18.8	19.3	19.5	17.4	16.1	13.0	11.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.3	12.9	14.5	17.0	20.4	23.3	23.2	24.1	21.5	19.7	16.2	13.8
		MCDB	14.3	15.4	19.8	24.5	27.8	30.7	31.5	31.9	27.6	23.8	18.2	14.7
	2%	WB	12.2	11.8	13.4	15.6	18.9	21.7	22.0	22.6	20.1	18.5	15.1	12.9
		MCDB	13.3	13.9	18.2	22.5	25.5	29.5	30.3	29.9	25.7	22.2	16.7	13.9
	5%	WB	11.2	11.1	12.5	14.3	17.6	20.5	21.0	21.5	19.1	17.5	14.1	12.1
		MCDB	12.4	13.1	16.3	19.8	23.4	27.2	28.4	28.1	24.5	20.8	15.8	13.2
	10%	WB	10.4	10.3	11.6	13.1	16.4	19.3	20.1	20.4	18.2	16.6	13.1	11.2
		MCDB	11.5	12.1	14.7	17.5	21.5	25.3	26.5	26.2	23.0	19.7	14.8	12.3
Mean Daily Temperature Range	5% DB	MDBR	6.3	7.9	9.4	10.2	10.8	11.0	11.4	11.0	9.2	7.3	6.6	
		MCDBR	6.5	10.3	13.3	14.2	14.2	14.5	14.4	15.4	14.4	11.0	8.2	6.9
		MCWBR	4.9	6.6	7.2	7.0	6.6	6.2	5.6	5.9	6.1	5.9	5.4	5.0
	5% WB	MCDBR	5.7	8.1	10.5	12.6	12.5	13.0	12.9	13.0	11.9	9.3	7.3	5.9
		MCWBR	4.7	5.7	6.1	6.5	6.4	6.1	5.5	5.7	5.8	5.4	5.3	4.8
Clear-Sky Solar Irradiance	taub	0.326	0.339	0.371	0.396	0.401	0.414	0.403	0.397	0.369	0.366	0.340	0.322	
	taud	2.460	2.408	2.320	2.250	2.291	2.288	2.316	2.352	2.407	2.428	2.446	2.455	
	Ebn at Noon	778	836	853	861	868	855	860	852	847	793	749	738	
	Edh at Noon	70	88	110	129	128	129	124	115	100	86	70	64	
All-Sky Solar Radiation	RadAvg	1.24	2.20	3.39	4.72	5.50	6.17	6.00	5.32	4.24	2.60	1.49	1.11	
	RadStd	0.13	0.24	0.40	0.46	0.43	0.45	0.32	0.36	0.28	0.22	0.17	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page